

CLOUD SIEM DETECTION AND INCIDENT WORKFLOW WITH MICROSOFT SENTINEL

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Advanced Security Operations
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Executive Summary

This laboratory exercise demonstrated the deployment, configuration, and operational use of Microsoft Sentinel as a cloud-native Security Information and Event Management (SIEM) solution within the Microsoft Azure environment. The objective of the exercise was to establish a dedicated security monitoring workspace, generate authorised Azure control-plane activities, investigate the resulting security events, and validate automated threat detection through Microsoft Sentinel analytics.

The implementation began with the creation of a dedicated Azure Resource Group, followed by the deployment of a Log Analytics Workspace and the onboarding of Microsoft Sentinel. Azure Activity Logs were connected to Microsoft Sentinel through the Azure Activity data connector, enabling the collection and centralised storage of Azure management events for security monitoring and analysis.

Authorised Azure management operations were generated using a laboratory script to simulate administrative control-plane activities, including resource creation, modification, and deletion. These activities produced Azure Activity logs that were ingested into the Log Analytics Workspace and analysed using Kusto Query Language (KQL). Multiple investigations were performed to examine management operations, identify failed activities, analyse resource deletions, evaluate ingestion latency, and correlate related events using Correlation IDs.

To validate the monitoring capability, a scheduled Microsoft Sentinel analytics rule was developed to detect resource deletion events within the laboratory environment. The rule successfully generated security alerts and incidents following the authorised deletion of a temporary Network Security Group. The resulting incident was investigated by reviewing alert metadata, correlating Azure Activity logs, analysing associated entities, and documenting the complete sequence of events from resource deletion to alert generation. The investigation confirmed that the detected activity originated from

an authorised laboratory exercise and demonstrated the effectiveness of Microsoft Sentinel in identifying administrative operations that could indicate malicious or unauthorised behaviour in a production environment.

Finally, all laboratory resources were securely removed, cost verification was performed, and the Azure environment was returned to its original state. Overall, the laboratory provided practical experience in deploying Microsoft Sentinel, configuring Azure log collection, performing KQL-based security investigations, developing scheduled analytics rules, triaging security incidents, and applying standard Security Operations Centre (SOC) investigation procedures within a Microsoft Azure cloud environment.

Tools Used

- **Microsoft Azure Portal:** For deploying and managing Azure resources.
- **Microsoft Sentinel:** For security monitoring, threat detection, and incident management.
- **Azure Log Analytics Workspace:** For collecting and storing Azure Activity logs.
- **Azure Activity Logs:** For monitoring Azure management operations.
- **Azure Cloud Shell (Bash):** For executing laboratory scripts and Azure CLI commands.
- **Azure CLI:** For creating, managing, and deleting Azure resources.
- **Kusto Query Language (KQL):** For querying and analysing security logs.

Learning Objectives

- Create a Log Analytics workspace and enable Microsoft Sentinel.
- Connect Azure Activity through the current supported workflow and verify ingestion.
- Use KQL to analyse control-plane events and entity context.
- Generate safe, reversible Azure activity without exposing a workload.
- Create and validate a scheduled analytics rule that returns TimeGenerated.
- Triage an incident, document evidence and complete cost-safe cleanup.

Methodology

Create the Dedicated Workspace

The first stage of the laboratory involved creating a dedicated Azure Resource Group to logically organise all cloud resources required for the Microsoft Sentinel deployment. A unique naming convention was used in accordance with the laboratory instructions to ensure resource isolation and simplify resource cleanup after completing the exercise. The following configuration was used during the resource group creation:

- Subscription: Azure for Students
- Resource Group: rg-csa303-2026-ASA-15643
- Region: South Africa North

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

asa2615643@icdfa.edu... INTERNATIONAL CYBERSECURITY...

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription *

Resource group name *

Region *

+ Create Manage view Refresh Export to CSV Open query Assign tags Add to service group Group by none

Filter for any field... Subscription equals all Location equals all Add filter

☐	Name ↑	Subscription	Location
☐	rg-csa303-2026-ASA-15643	Azure for Students	South Africa North

Azure Portal showing the Create Resource Group page with the configured values.

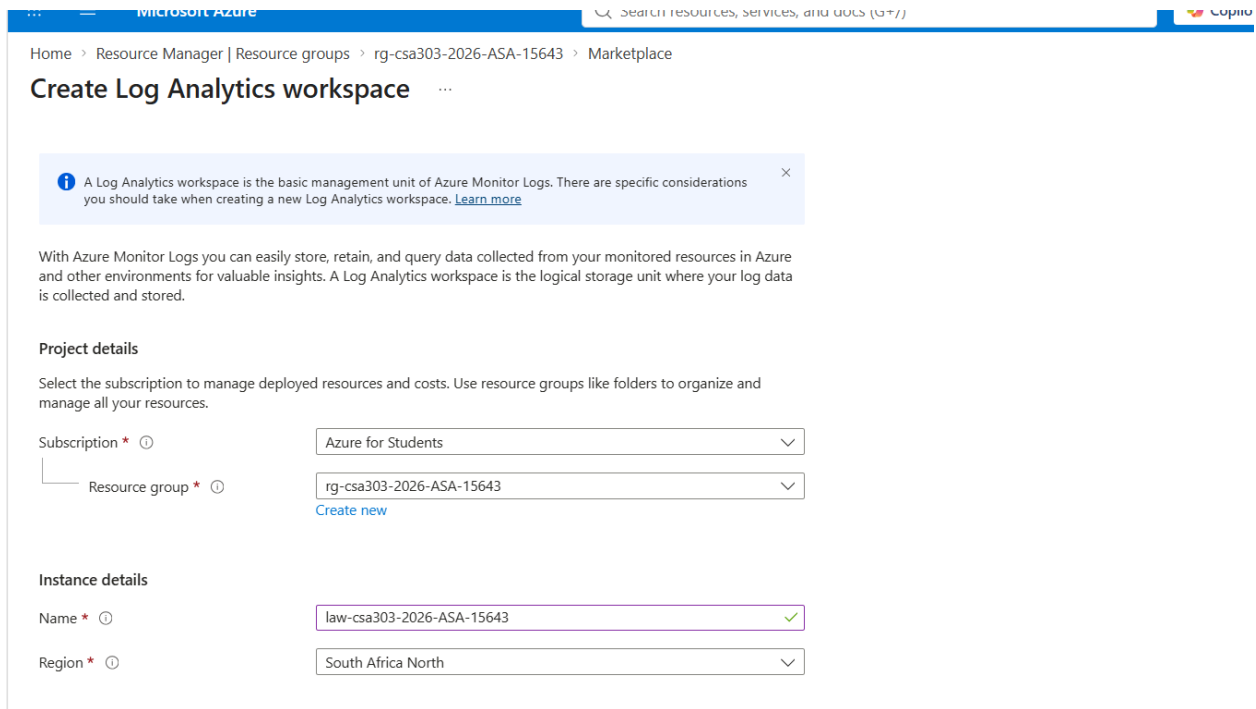
Creation of the Log Analytics Workspace

After creating the resource group, a Log Analytics Workspace was deployed within the resource group to serve as the central repository for log collection and security monitoring.

The following configuration was specified:

- **Workspace Name:** law-csa303-2026-ASA-15643
- **Subscription:** Azure for Students
- **Resource Group:** rg-csa303-2026-ASA-15643
- **Region:** South Africa North

The workspace deployment completed successfully without any errors.



The screenshot displays the 'Create Log Analytics workspace' page in the Microsoft Azure portal. The breadcrumb navigation at the top reads: Home > Resource Manager | Resource groups > rg-csa303-2026-ASA-15643 > Marketplace. The page title is 'Create Log Analytics workspace'. A blue information banner at the top states: 'A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)'. Below this, a descriptive paragraph explains that with Azure Monitor Logs, data can be stored, retained, and queried from monitored resources. The 'Project details' section instructs the user to select a subscription and resource group. The 'Subscription' dropdown is set to 'Azure for Students' and the 'Resource group' dropdown is set to 'rg-csa303-2026-ASA-15643', with a 'Create new' link below the resource group dropdown. The 'Instance details' section contains the 'Name' field, which is populated with 'law-csa303-2026-ASA-15643' and has a green checkmark, and the 'Region' dropdown, which is set to 'South Africa North'.

Microsoft Azure

SEARCH RESOURCES, SERVICES, AND DOCS (0+7)

Home > Resource Manager | Resource groups > rg-csa303-2026-ASA-15643 > Marketplace

Create Log Analytics workspace

i A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure for Students ▼

Resource group * ⓘ rg-csa303-2026-ASA-15643 ▼
[Create new](#)

Instance details

Name * ⓘ law-csa303-2026-ASA-15643 ✓

Region * ⓘ South Africa North ▼

Create Log Analytics workspace

Validation passed

Basics

Tags

Review + Create

Log Analytics workspace

by Microsoft

Basics

Subscription

Resource group

Name

Region

Pricing

Pricing tier

The cost of your workspace depends on the volume of data ingested and how long it is retained. Regional pricing details are available on the [Azure Monitor pricing page](#). You can change to a different pricing tier after the workspace is created. [Learn more](#) about Log Analytics pricing models.

Tags

None

Delete

Cancel

Redeploy

Download

Refresh

✓ Your deployment is complete

Deployment name : Microsoft.LogAnalyticsOMS

Subscription : [Azure for Students](#)

Resource group : [rg-csa303-2026-ASA-15643](#)

Start time : 8/4/2026, 11:45:31 AM

Correlation ID : 2abfcddf-d2aa-48c1-b7fc-8776182fef46

Deployment details

Resource	Type	Status	Operation details
✓ law-csa303-2026-ASA-15643	Log Analytics workspace	OK	Operation details

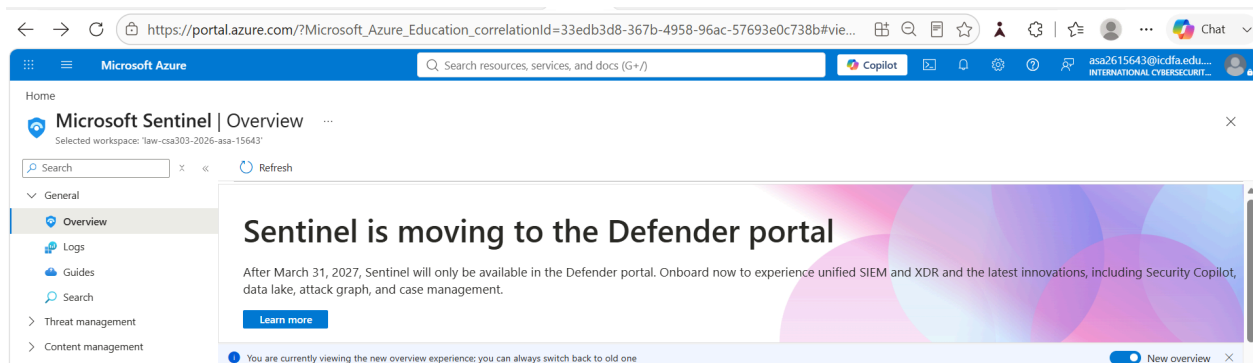
Next steps

Go to resource

Deployment completed notification successfully.

Microsoft Sentinel Deployment

Microsoft Sentinel was enabled using the newly created Log Analytics Workspace. The workspace was selected during the onboarding process, allowing Sentinel to begin operating as the Security Information and Event Management (SIEM) platform for the laboratory environment.



Recording Workspace Information

The workspace details required for the laboratory report were documented. Sensitive information such as the Subscription ID was partially masked in accordance with security best practices.

The following information was recorded:

- Subscription ID (last six characters only)
- Resource Group Name
- Workspace Name
- Azure Region
- Workspace Resource ID

! Azure Monitor will enforce TLS versions 1.2 and above starting March 1, 2026. To avoid data loss and service disruption please **upgrade to AMA** and ensure your **OS supports TLS 1.2** or higher. [Learn More About TLS](#). [Learn More About Agents](#).

^ Essentials

Resource group [\(move\)](#) : [rg-csa303-2026-asa-15643](#)

Status : Active

Location : South Africa North

Subscription [\(move\)](#) : [Azure for Students](#)

Subscription ID : 56f22d6e-541f-4bf2-8c95-5a12c160f4c4

Tags [\(edit\)](#) : [Add tags](#)

Workspace Name : law-csa303-2026-ASA-15643

Workspace ID : 5ebba9fe-e0c5-41e9-8b1b-9b1d06d561af

Pricing tier : Pay-as-you-go

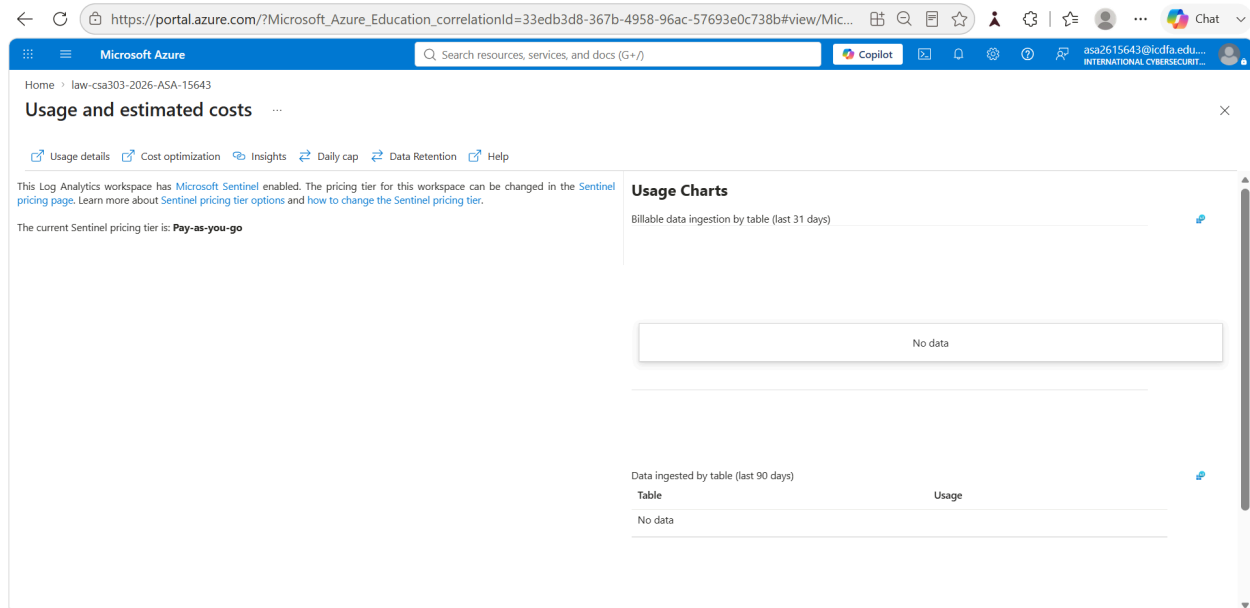
Access control mode : Use resource or workspace permissions

Operational issues : [OK](#)

[JSON View](#)

Usage and Cost Baseline

Before ingesting any security logs, the Usage and Estimated Costs page of the Log Analytics Workspace was reviewed to establish the initial operational baseline. This baseline will later be used to compare resource utilisation after completing the laboratory activities.



Connect Azure Activity

Install the Azure Activity Solution

The Microsoft Sentinel Content Hub was accessed to install the Azure Activity solution. This solution provides the required data connector for collecting Azure control-plane events, enabling Microsoft Sentinel to monitor management operations performed within the Azure subscription.

The Content Hub was searched for Azure Activity, and the solution was installed into the existing Microsoft Sentinel workspace.

☐	Content title	Status	Content source
☐	 Azure Activity FEATURED	 Installed	Solution

Configure the Azure Activity Data Connector

After installing the Azure Activity solution, the Data Connectors section of Microsoft Sentinel was opened. The Azure Activity connector was selected, and the configuration wizard was launched to enable ingestion of Azure Activity logs from the lab subscription.

Only the assigned Azure for Students subscription was selected, in accordance with the laboratory requirements.



Azure Activity

Connected

Status



Microsoft

Provider

13 minutes ago

Last Log Received

Description

Azure Activity Log is a subscription log that provides insight into subscription-level events that occur in Azure, including events from Azure Resource Manager operational data, service health events, write operations taken on the resources in your subscription, and the status of activities performed in Azure.

Last data received

08/04/26, 02:31 PM

Content source

Azure Activity

Version

2.0.0

Author

Microsoft

Supported by

Microsoft Corporation | Email

Related content

Instructions



Prerequisites

To integrate with Azure Activity make sure you have:

✓

Workspace: read and write permissions.

!

Policy: owner role assigned for each policy assignment scope.

!

Subscription: owner role permission on the relevant subscription



Configuration

!

This connector has been updated to use the diagnostics settings back-end pipeline, which provides increased functionality. Connectors using this pipeline can also be governed at scale by Azure Policy. Learn more about the new [Azure Activity connector](#). Follow the instructions below to upgrade your connector to the diagnostics settings pipeline.

Validate Azure Activity Ingestion

Following connector configuration, sufficient time was allowed for Azure Activity events to begin flowing into the Log Analytics Workspace. The validation query supplied in the laboratory manual was executed to verify successful ingestion of Azure Activity records.

```
AzureActivity
| where TimeGenerated > ago(24h)
| summarize Events=count(),
             First=min(TimeGenerated),
             Last=max(TimeGenerated),
             Operations=dcount(OperationNameValue)
| project Events, First, Last, Operations
```

The screenshot shows the Azure Log Analytics query interface. At the top, there's a 'Run' button and a 'Time range: Set in query' dropdown. Below the query editor, the query is displayed: `1 AzureActivity`, `2 | where TimeGenerated > ago(24h)`, `3 | summarize Events=count(), First=min(TimeGenerated), Last=max(TimeGenerated),`, `4 Operations=dcount(OperationNameValue)`, `5 | project Events, First, Last, Operations`. The results section shows a table with columns: Events, First [UTC], Last [UTC], and Operations. The results are: Events: 8, First [UTC]: 2026-08-04T13:30:59.824487Z, Last [UTC]: 2026-08-04T13:35:07.0124731Z, Operations: 3.

Events	First [UTC]	Last [UTC]	Operations
8	2026-08-04T13:30:59.824487Z	2026-08-04T13:35:07.0124731Z	3

Column	Meaning	Your Result
Events	Total number of Azure Activity log events received	8
First [UTC]	The timestamp of the earliest event found in the selected time range	2026-08-04T13:30:59.824487Z
Last [UTC]	The timestamp of the most recent event found in the selected time range	2026-08-04T13:35:07.0124731Z
Operations	Number of unique Azure operations recorded	3

Generate Validation Activity (If Required)

After the Azure Policy assignment and remediation task completed successfully, the Azure Activity connector status changed to Connected. A validation query was executed in the Log Analytics workspace to confirm that Azure Activity logs were being ingested successfully. The query returned 8 Azure Activity events representing 3 unique management operations, confirming successful log collection and ingestion into Microsoft Sentinel.

Generate Authorised Control-Plane Events

The objective of this phase was to generate authorised Azure control-plane events that would be captured by the Azure Activity data connector and ingested into the Log Analytics Workspace. These events establish a controlled dataset for subsequent monitoring, investigation, and correlation activities using Microsoft Sentinel.

Azure Cloud Shell (Bash) was launched from the Azure portal to provide a command-line environment for executing the laboratory script. The script was executed using the assigned student identifier and the approved Azure deployment region.

```
chmod +x generate-azure-activity-events.sh
./generate-azure-activity-events.sh \
--student-id 2026/ASA/15643
--location southafricanorth
```

```
theophilus [ ~ ]$ ls
generate-azure-activity-events.sh
theophilus [ ~ ]$ chmod +x generate-azure-activity-events.sh
theophilus [ ~ ]$ ls
generate-azure-activity-events.sh
theophilus [ ~ ]$ ./generate-azure-activity-events.sh \
--student-id ASA-15643 \
--location southafricanorth
CSA303_START UTC-2026-08-04T14:36:48Z
Subscription      User      Tenant
-----
56f22d6e-541f-4bf2-8c95-5a12c160f4c4 asa2615643@icdfa.edu.ng 8616658e-4dc7-47eb-8a89-67cd1a74ebd4
Location      Name
-----
southafricanorth rg-csa303-asa-15643

Access      Description      DestinationAddressPrefix      DestinationPortRange      Direction      Name      Priority      Protocol      ProvisioningState      Res
SourceGroup      SourceAddressPrefix      SourcePortRange
-----
Allow      CSA303 unattached audit-only rule; remove during cleanup *      3389      Inbound      CSA303-Temporary-RDP-Audit      4090      Tcp      Succeeded      rg-
csa303-asa-15643      *
Location      Name
-----
southafricanorth rg-csa303-asa-15643
(SubscriptionNotFound) Subscription 56f22d6e-541f-4bf2-8c95-5a12c160f4c4 was not found.
Code: SubscriptionNotFound
Message: Subscription 56f22d6e-541f-4bf2-8c95-5a12c160f4c4 was not found.
Storage account creation was denied or unavailable. Continue with permitted events and document the control.
CSA303_END UTC-2026-08-04T14:37:08Z
Resource group: rg-csa303-asa-15643
NSG: nsg-csa303-asa-15643 (must remain unattached and be deleted)
Run cleanup-section1-1b.sh after evidence capture
```

Verify the Laboratory Resource Group

After executing the event generation script, it was confirmed that only the designated laboratory resource group was used during the exercise. The script verified the existence of the resource group and reused it rather than creating additional resource groups. The expected resource group was: rg-csa303-asa-15643

This verification ensures that all authorised control-plane activities were performed within the designated laboratory environment without affecting resources outside the assigned resource group.

Subscription	User	Tenant
66f22d6e-541f-4bf2-8c95-5a12c160f4c4	asa2615643@icdfa.edu.ng	8616658e-4dc7-47eb-8a89-67cd1a74ebd4
Location	Name	
southafricanorth	rg-csa303-asa-15643	

Verify the Network Security Group

The script successfully created a Network Security Group (NSG) named:
nsg-csa303-asa-15643

The NSG was created solely for the purpose of generating Azure Activity logs. It was confirmed that the NSG was not associated with any subnet or network interface, ensuring that it did not influence network traffic or expose any Azure resources.

Leaving the NSG unattached ensured that the activity remained a management-plane operation for monitoring purposes only.

```
NSG: nsg-csa303-asa-15643 (must remain unattached and be deleted)
Run cleanup-sentinel-lab.sh after evidence capture.
theophilus [ ~ ]$
```

Verify the Temporary Security Rule

As part of the authorised event generation process, the script created a temporary inbound Network Security Group rule permitting TCP port 3389 (Remote Desktop Protocol).

The purpose of this rule was not to allow remote access but to generate Azure management-plane events that could later be detected and investigated within Microsoft Sentinel.

The temporary rule remained unattached to any Azure subnet or network interface and therefore did not expose any Azure resource to inbound traffic. The rule was designed to be removed automatically during the cleanup phase.

Access	Description	DestinationAddressPrefix	DestinationPortRange	Direction	Name	Priority	Protocol	ProvisioningState	ResourceGroup	SourceAddressPrefix	SourcePortRange
Allow	CSA383 unattached audit-only rule; remove during cleanup	*	3389	Inbound	CSA383-Temporary-RDP-Audit	4898	Tcp	Succeeded	rg-csa383-asa-15643	*	*

Verify Storage Account Operations

The script attempted to create a uniquely named Azure Storage Account, apply resource tags, and subsequently delete the storage account. These actions were intended to generate additional Azure Activity events for monitoring and investigation.

During execution, the storage account creation was prevented by the subscription's organisational policy, resulting in the following message:

"Storage account creation was denied or unavailable. Continue with permitted events and document the control."

In accordance with the laboratory instructions, the policy restriction was not bypassed or modified. Instead, the remaining authorised control-plane activities continued successfully while the policy restriction was documented.

```
Message: Subscription 5072200c-541f-4b72-bc55-5d12c10074c7 was not found.  
Storage account creation was denied or unavailable. Continue with permitted events and document the control.
```

Record the Evidence Collection Window

Upon completion, the script displayed the evidence collection window, identifying the exact period during which the authorised Azure Activity events were generated.

Parameter	Value
Start UTC	2026-08-04T14:29:00Z
End UTC	2026-08-04T14:31:16Z

The complete terminal output was preserved as evidence of successful execution and will be used during the investigation and correlation phases of the laboratory.

```
--LOCATION SUBSCRIPTION  
CSA303_START_UTC=2026-08-04T14:36:48Z
```

```
CSA303_END_UTC=2026-08-04T14:37:08Z
```

KQL Investigation

Activity in the Laboratory Resource Group

To investigate the administrative activities performed within the laboratory environment, the Logs page in Microsoft Sentinel was opened from the Azure portal. The Kusto Query Language (KQL) editor was used to query the AzureActivity table, which contains Azure Resource Manager (ARM) management-plane events. The following query was entered into the query editor and executed by selecting Run. The query was configured to retrieve activity generated within the last 48 hours for the designated laboratory resource group.

```

let LabRG = "rg-csa303-asa-15643";
AzureActivity
| where TimeGenerated > ago(48h)
| where ResourceGroup =~ LabRG
| project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue,
    ResourceProviderValue, ResourceId, CorrelationId, _ResourceId
| order by TimeGenerated asc

```

Query Explanation

- **Table:** The AzureActivity table was queried because it stores Azure Resource Manager (ARM) management operations, including resource creation, updates, deletions, and policy enforcement events.
- **Filter:** The where statements limited the results to the last 48 hours and to events associated with the laboratory resource group (rg-csa303-asa-15643).
- **Projection:** The project operator displayed only the fields required for the investigation, including the event time, initiating user, operation name, status, resource details, and Correlation ID.
- **Aggregation:** No aggregation was performed in this query. Instead, the results were sorted in chronological order using order by to reconstruct the sequence of administrative activities.

▶ Run | ▼

Time range : Set in query

Show : 1000 results

KQL mode ▼

```
1 let LabRG = "rg-csa303-asa-15643";
2 AzureActivity
3 | where TimeGenerated > ago(48h)
4 | where ResourceGroup =~ LabRG
5 | project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue,
6 |   ResourceProviderValue, ResourceId, CorrelationId, _ResourceId
7 | order by TimeGenerated asc
```

Results

Chart

🔖

Add bookmark

☐	TimeGenerated [UTC]	Caller	OperationNameValue	ActivityStatusValue	ResourceProviderValue	CorrelationId
☐	> 8/4/2026, 2:29:10.137 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/WRITE	Failure	MICROSOFT.NETWORK	c83c8af6-221b-47
☐	> 8/4/2026, 2:30:57.075 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/WRITE	Start	MICROSOFT.NETWORK	a4be5364-c686-4C
☐	> 8/4/2026, 2:31:03.841 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/WRITE	Accept	MICROSOFT.NETWORK	a4be5364-c686-40
☐	> 8/4/2026, 2:31:07.282 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/SECU...	Start	MICROSOFT.NETWORK	8fd5138b-a5e9-47
☐	> 8/4/2026, 2:31:07.422 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/SECU...	Accept	MICROSOFT.NETWORK	8fd5138b-a5e9-47
☐	> 8/4/2026, 2:31:09.144 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/WRITE	Success	MICROSOFT.NETWORK	a4be5364-c686-4C

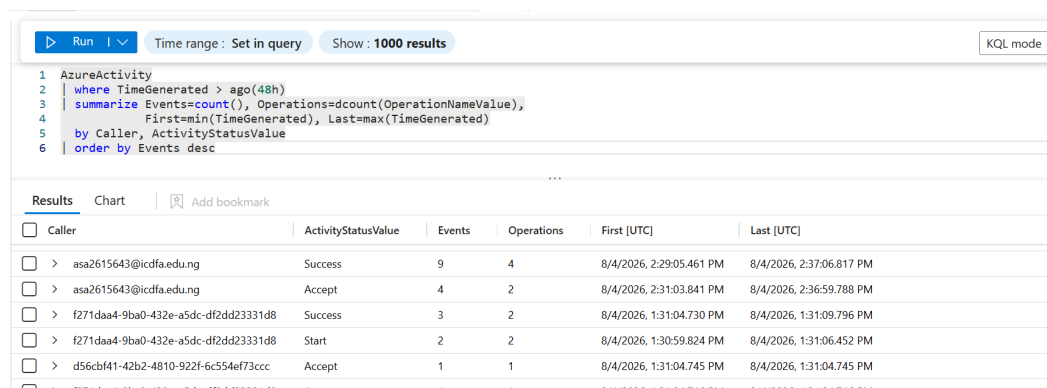
Operations by Caller and Result

To identify the Azure identities responsible for administrative activities and evaluate the outcome of each operation, the Logs page in Microsoft Sentinel was used to query the AzureActivity table. The following KQL query was entered into the query editor and executed by selecting Run. The query summarises all management operations performed within the last 48 hours, grouping the results by the initiating user (Caller) and the operation status.

```
AzureActivity
| where TimeGenerated > ago(48h)
| summarize Events=count(), Operations=dcount(OperationNameValue),
             First=min(TimeGenerated), Last=max(TimeGenerated)
  by Caller, ActivityStatusValue
| order by Events desc
```

Query Explanation

- Table: The AzureActivity table was queried to analyse management-plane operations performed within the Azure subscription.
- Filter: The where statement limits the results to activities generated within the last 48 hours.
- Projection: No project operator was used because the query summarises the data instead of displaying individual event records.
- Aggregation: The summarise operator groups events by Caller and ActivityStatusValue, calculating the total number of events, the number of unique operations performed, and the first and last recorded event for each user.



The screenshot shows the Microsoft Sentinel query editor with a KQL query and its results. The query is as follows:

```
1 AzureActivity
2 | where TimeGenerated > ago(48h)
3 | summarize Events=count(), Operations=dcount(OperationNameValue),
4             First=min(TimeGenerated), Last=max(TimeGenerated)
5   by Caller, ActivityStatusValue
6 | order by Events desc
```

The results table displays the following data:

Caller	ActivityStatusValue	Events	Operations	First [UTC]	Last [UTC]
> asa2615643@icdfa.edu.ng	Success	9	4	8/4/2026, 2:29:05.461 PM	8/4/2026, 2:37:06.817 PM
> asa2615643@icdfa.edu.ng	Accept	4	2	8/4/2026, 2:31:03.841 PM	8/4/2026, 2:36:59.788 PM
> f271daa4-9ba0-432e-a5dc-df2dd23331d8	Success	3	2	8/4/2026, 1:31:04.730 PM	8/4/2026, 1:31:09.796 PM
> f271daa4-9ba0-432e-a5dc-df2dd23331d8	Start	2	2	8/4/2026, 1:30:59.824 PM	8/4/2026, 1:31:06.452 PM
> d56cbf41-42b2-4810-922f-6c554ef73ccc	Accept	1	1	8/4/2026, 1:31:04.745 PM	8/4/2026, 1:31:04.745 PM
> f271daa4-9ba0-432e-a5dc-df2dd23331d8	Accept	1	1	8/4/2026, 1:31:04.746 PM	8/4/2026, 1:31:04.746 PM

Resource Deletions

To identify resources removed during the laboratory exercise, the Logs page in Microsoft Sentinel was used to query the AzureActivity table. The following query filters the activity logs to display only resource deletion operations performed within the last 48 hours.

```
AzureActivity
| where TimeGenerated > ago(48h)
| where OperationNameValue endswith "/DELETE"
| project TimeGenerated, Caller, OperationNameValue, ResourceGroup,
    ResourceId, ActivityStatusValue, CorrelationId
| order by TimeGenerated desc
```

Query Explanation

- Table: The AzureActivity table was queried because it records Azure management operations, including resource deletion events.
- Filter: The query filters events generated within the last 48 hours and selects only operations whose names end with /DELETE.
- Projection: The project operator displays the event timestamp, initiating user, deleted resource, operation status, resource group, and Correlation ID required for the investigation.
- Aggregation: No aggregation was performed. The results were sorted in descending order to display the most recent deletion events first.

The screenshot shows the Microsoft Sentinel KQL query interface. At the top, there are buttons for 'Run', 'Time range: Set in query', and 'Show: 1000 results'. A dropdown menu on the right is set to 'KQL mode'. The query editor contains the following KQL query:

```
1 AzureActivity
2 | where TimeGenerated > ago(48h)
3 | where OperationNameValue endswith "/DELETE"
4 | project TimeGenerated, Caller, OperationNameValue, ResourceGroup,
5   ResourceId, ActivityStatusValue, CorrelationId
6 | order by TimeGenerated desc
```

Below the query editor, there are tabs for 'Results' and 'Chart'. A message box at the bottom states: 'No results found from the specified time range. Try [selecting another time range](#). [Review the query details for more information.](#)'

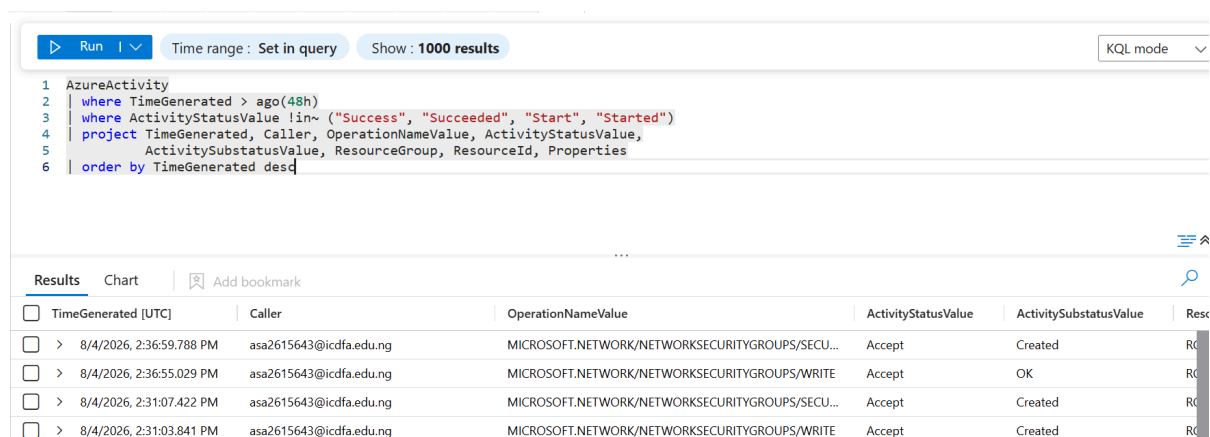
Failed or Denied Operations

To investigate unsuccessful administrative activities, the Logs page in Microsoft Sentinel was used to query the AzureActivity table. The following KQL query retrieves management operations that did not complete successfully, allowing policy violations, permission issues, and failed deployments to be examined.

```
AzureActivity
| where TimeGenerated > ago(48h)
| where ActivityStatusValue !in~ ("Success", "Succeeded", "Start", "Started")
| project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue,
    ActivitySubstatusValue, ResourceGroup, ResourceId, Properties
| order by TimeGenerated desc
```

Query Explanation

- **Table:** The AzureActivity table was queried to analyse unsuccessful Azure Resource Manager operations.
- **Filter:** The query limits the results to the last 48 hours and excludes successful or in-progress operations, returning only failed or denied activities.
- **Projection:** The project operator displays relevant fields including the event time, initiating user, operation name, operation status, sub-status, affected resource group, resource identifier, and additional properties describing the failure.
- **Aggregation:** No aggregation was performed. The results were sorted in descending order to prioritise the most recent failed or denied operations.



The screenshot shows the Microsoft Sentinel KQL query interface. At the top, there are buttons for 'Run', 'Time range: Set in query', and 'Show: 1000 results'. The KQL query is displayed in a text area, and below it, the 'Results' tab is active, showing a table of query results. The table has columns for TimeGenerated [UTC], Caller, OperationNameValue, ActivityStatusValue, ActivitySubstatusValue, and ResourceId. The results show four entries, all with 'Accept' status and 'Created' sub-status, indicating successful operations.

	TimeGenerated [UTC]	Caller	OperationNameValue	ActivityStatusValue	ActivitySubstatusValue	ResourceId
☐	> 8/4/2026, 2:36:59.788 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/SECU...	Accept	Created	RQ...
☐	> 8/4/2026, 2:36:55.029 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/WRITE	Accept	OK	RQ...
☐	> 8/4/2026, 2:31:07.422 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/SECU...	Accept	Created	RQ...
☐	> 8/4/2026, 2:31:03.841 PM	asa2615643@icdfa.edu.ng	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/WRITE	Accept	Created	RQ...

Ingestion Delay Estimate

To estimate the time taken for Azure Activity logs to become available in Microsoft Sentinel, the Logs page in Microsoft Sentinel was opened and the following Kusto Query Language (KQL) query was entered into the query editor. The query calculates the difference between the event generation time and the log ingestion time for Azure Activity events recorded within the last 48 hours.

```
AzureActivity  
| where TimeGenerated > ago(48h)  
| extend IngestionDelay = ingestion_time() - TimeGenerated  
| summarize Events=count(), AvgDelay=avg(IngestionDelay),  
              P95Delay=percentile(IngestionDelay,95), MaxDelay=max(IngestionDelay)
```

Query Explanation

Table: The AzureActivity table was queried because it contains Azure Resource Manager (ARM) management events collected from the Azure subscription.

Filter: The where statement restricts the query to Azure Activity events generated within the last 48 hours, ensuring that only recent events are analysed.

Projection: No project operator is used in this query. Instead, the extend operator creates a new field named IngestionDelay, which calculates the time difference between when an event occurred (TimeGenerated) and when it became available for querying (ingestion_time()).

Aggregation: The summarise operator calculates:

- Events – the total number of Azure Activity events analysed.
- AvgDelay – the average log ingestion delay.
- P95Delay – the 95th percentile ingestion delay, indicating the delay experienced by most events.
- MaxDelay – the maximum ingestion delay observed during the investigation.

▶ Run | ▼

Time range: Set in query

Show: 1000 results

KQL mode

```
1 AzureActivity
2 | where TimeGenerated > ago(48h)
3 | extend IngestionDelay = ingestion_time() - TimeGenerated
4 | summarize Events=count(), AvgDelay=avg(IngestionDelay),
5 | P95Delay=percentile(IngestionDelay,95), MaxDelay=max(IngestionDelay)
```

Results | Chart | Add bookmark

☐ Events

☐ AvgDelay

☐ P95Delay

☐ MaxDelay

>

32

00:06:06.8257994

00:11:13.6875944

00:11:18.5938510

Chronological Timeline of Events

S/N	Time (UTC)	Operation Name	Activity Status	Correlation ID	Description
1	2026-08-04 2:29:02	MICROSOFT .RESOURCE S/SUBSCRIPTIONS/RESOURCEGROUPS/WRITE	Start	f088fd3f-c38c-4bdf-9f14-fb77e9156bae	Azure initiated the resource group write operation.
2	2026-08-04 2:29:05	MICROSOFT .RESOURCE S/SUBSCRIPTIONS/RESOURCEGROUPS/WRITE	Success	f088fd3f-c38c-4bdf-9f14-fb77e9156bae	The resource group was successfully created or reused.
3	2026-08-04 2:29:09	MICROSOFT .NETWORK/NETWORKSECURITYGR	Start	c83c8af6-221b-4720-ba0b-6f9c3266958f	Azure initiated the Network Security

		OUPS/WRIT E			Group (NSG) creation.
4	2026-08-04 2:29:10	MICROSOFT .NETWORK/ NETWORKS ECURITYGR OUPS/WRIT E	Failure	c83c8af6-22 1b-4720-ba0 b-6f9c3266 958f	The initial NSG creation attempt failed.
5	2026-08-04 2:30:57	MICROSOFT .NETWORK/ NETWORKS ECURITYGR OUPS/WRIT E	Start	a4be5364-c 686-4017-8 23c-581339 98c5f1	Azure retrieved the NSG creation operation.
6	2026-08-04 2:31:03	MICROSOFT .NETWORK/ NETWORKS ECURITYGR OUPS/WRIT E	Accept	a4be5364-c 686-4017-8 23c-581339 98c5f1	Azure accepted the NSG creation request.
7	2026-08-04 2:31:07	MICROSOFT .NETWORK/ NETWORKS ECURITYGR OUPS/SECU RITYRULES/ WRITE	Start	8fd5138b-a 5e9-4777-9f 74-4209e60 9c4cd	Azure initiated creation of the temporary RDP security rule.
8	2026-08-04 2:31:07	MICROSOFT .NETWORK/ NETWORKS ECURITYGR OUPS/SECU RITYRULES/ WRITE	Accept	8fd5138b-a 5e9-4777-9f 74-4209e60 9c4cd	Azure accepted the creation of the temporary RDP security rule.

The following timeline was reconstructed from the Azure Activity logs generated during the execution of the authorised event generation script.

CorrelationId Analysis

The CorrelationId was used to group related Azure Activity records generated during the execution of the authorised control-plane operations. This field uniquely identifies all log records associated with a single Azure Resource Manager (ARM) request, allowing security analysts to reconstruct the complete sequence of an administrative action. During the investigation, several Correlation IDs were observed:

Correlation ID	Operation	Observed Activity Status
f088fd3f-c38c-4bdf-9f14-fb77e9156bae	Resource Group Write	Start - Success
c83c8af6-221b-4720-ba0b-6f9c3266958f	Network Security Group (NSG) Write – Initial Attempt	Start - Failure
a4be5364-c686-4017-823c-58133998c5f1	Network Security Group (NSG) Write – Retry	Start - Accept
8fd5138b-a5e9-4777-9f74-4209e609c4cd	NSG Security Rule Write	Start - Accept

The

CorrelationId f088fd3f-c38c-4bdf-9f14-fb77e9156bae demonstrates how a single administrative action can generate multiple Azure Activity records. In this case, the resource group write operation produced both a Start event, indicating that Azure Resource Manager had begun processing the request, and a Success event, confirming that the operation completed successfully.

Similarly, the CorrelationId c83c8af6-221b-4720-ba0b-6f9c3266958f grouped the initial Network Security Group creation attempt, which generated a Start event followed by a Failure event. Azure subsequently retried the operation using a new CorrelationId (a4be5364-c686-4017-823c-58133998c5f1), resulting in Start and Accept events. This demonstrates that Azure may create separate correlation groups for subsequent attempts while preserving the complete audit trail of the deployment process.

From a SOC perspective, CorrelationId is an important investigation field because it links all records belonging to the same management operation. Rather than analysing

each event independently, analysts can group related records using the CorrelationId to reconstruct the full lifecycle of an administrative action, determine whether the operation completed successfully or failed, and identify the sequence of events leading to the outcome. This significantly improves incident investigation, troubleshooting, and forensic analysis.

Event Time vs Ingestion Time

The TimeGenerated field records when an Azure management event actually occurred, whereas ingestion_time() indicates when the event became available within the Log Analytics Workspace and Microsoft Sentinel.

The investigation identified a measurable ingestion delay between event generation and log availability. This delay can affect scheduled analytics rules configured to execute every five minutes. If log ingestion exceeds the rule execution interval, newly generated events may not be detected during the first execution cycle and instead be processed during the subsequent run. Consequently, SOC analysts should consider expected ingestion latency when configuring detection rules and alert thresholds to minimise delayed detections.

Incident Entity Mapping

The Caller field is the most appropriate incident entity for this investigation because it identifies the Azure user or service principal responsible for performing each management operation.

Mapping the Caller as an incident entity enables Microsoft Sentinel to associate multiple related events with the responsible identity, improving investigation efficiency, alert correlation, incident triage, and threat hunting activities. During an investigation, analysts can quickly determine who initiated an operation and assess whether the activity was authorised or potentially malicious.

Scheduled Analytics Rule

A Scheduled Analytics Rule was created in Microsoft Sentinel to detect Azure resource deletion events occurring within the CSA303 laboratory resource groups. The rule was configured using the supplied KQL query to monitor Azure Activity logs for DELETE operations. The rule was assigned a Medium severity level, configured to execute every 15 minutes with a 15-minute lookback period, and set to automatically create incidents upon detecting matching events. Entity mapping was configured to associate the initiating account (Caller) and the affected Azure resource (ResourceId) with generated alerts.

Analytics rule details	
Name	CSA303 – Resource Deletion in Lab Subscription – ASA-15643
Description	#INC_CORR# Detects Azure resource deletion events occurring in CSA303 laboratory resource groups.
MITRE ATT&CK	-
Severity	■ ■ ■ Medium
Status	◦ Enabled
Analytics rule settings	
Rule query	let LabRGPref = "rg-csa303-"; AzureActivity where TimeGenerated > ago(15m) where ResourceGroup startswith LabRGPref where OperationNameValue endswith "/DELETE" project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup, ResourceId, CorrelationId
Rule frequency	Run query every 15 minutes
Rule period	Last 15 minutes data
Rule start time	Automatic
Rule threshold	Trigger alert if query returns more than 0 results
Event grouping	Group all events into a single alert
Suppression	Not configured

Validation of the Analytics Rule

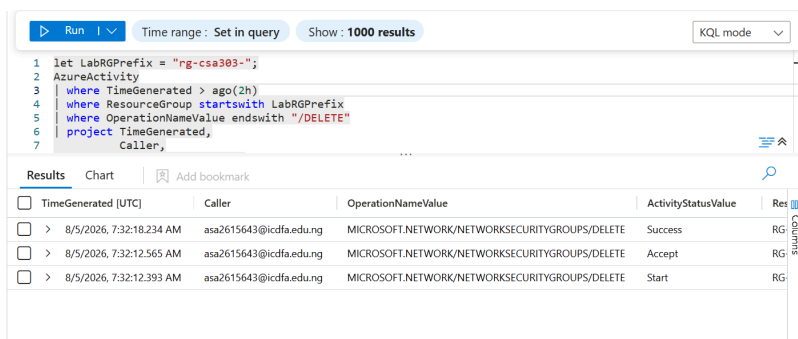
To validate the scheduled analytics rule, a second disposable Azure resource was created within the laboratory resource group rg-csa303-asa-15643. A temporary Network Security Group (NSG) named nsg-csa303-delete-test was created using Azure CLI and immediately deleted after successful deployment. This generated authorised Azure Activity DELETE events required to test the detection logic of the scheduled analytics rule without affecting production resources.

The environment was then left unchanged while allowing sufficient time for Azure Activity log ingestion and scheduled analytics rule execution. No additional resources were created or deleted during the waiting period to avoid generating unnecessary activity that could interfere with the validation process

```
theophilus [ ~ ]$ az network nsg delete \
--resource-group rg-csa303-asa-15643 \
--name nsg-csa303-delete-test
```

Verification of Azure Activity Logs

Following the waiting period, the Azure Activity table was queried in Microsoft Sentinel using the scheduled analytics rule query to verify that the Network Security Group deletion had been successfully ingested into the Log Analytics workspace. The query was executed over a two-hour time range to account for possible ingestion latency and to confirm that the Start, Accept, and Success stages of the deletion operation were recorded.



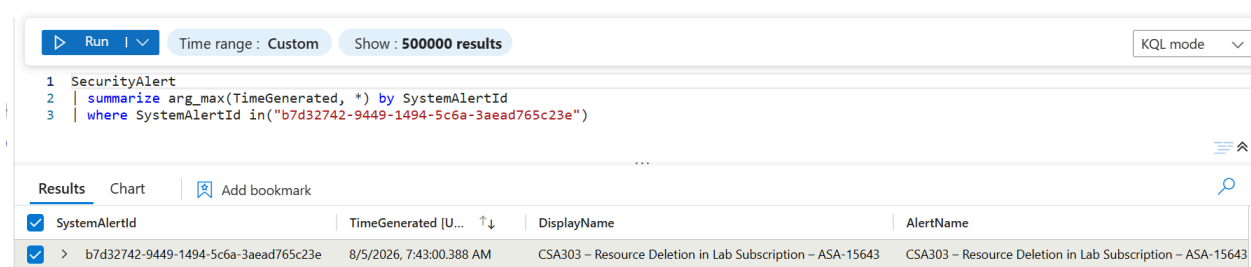
The screenshot shows a KQL query in Microsoft Sentinel. The query filters for AzureActivity events where the ResourceGroup starts with 'rg-csa303-' and the OperationNameValue ends with '/DELETE'. The results table displays three rows of data, each representing a stage of the NSG deletion process: Start, Accept, and Success.

TimeGenerated [UTC]	Caller	OperationNameValue	ActivityStatusValue	Res
> 8/5/2026, 7:32:18.234 AM	asa2615643@icdfa.edung	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/DELETE	Success	RG
> 8/5/2026, 7:32:12.565 AM	asa2615643@icdfa.edung	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/DELETE	Accept	RG
> 8/5/2026, 7:32:12.393 AM	asa2615643@icdfa.edung	MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/DELETE	Start	RG

Alert Validation and Detection Delay

After confirming that the Azure Activity DELETE events had been ingested into the Log Analytics workspace, Microsoft Sentinel was monitored to determine whether the scheduled analytics rule successfully generated an alert. The alert timestamp was compared with the timestamp of the successful Azure Activity DELETE event to determine the actual detection delay between the management operation and alert generation.

The Azure Activity logs recorded the successful deletion of the temporary Network Security Group at 07:32:18.234 UTC, while Microsoft Sentinel generated the corresponding alert at 07:43:00.388 UTC. The observed detection delay was therefore approximately 10 minutes and 42 seconds, which is consistent with Azure Activity log ingestion and the configured 15-minute analytics rule execution schedule.



The screenshot shows the Microsoft Sentinel interface with a KQL query executed. The query filters for a specific SecurityAlert ID and summarizes the maximum time generated. The results table shows one entry with the SystemAlertId, TimeGenerated, DisplayName, and AlertName.

SystemAlertId	TimeGenerated [UTC]	DisplayName	AlertName
b7d32742-9449-1494-5c6a-3aead765c23e	8/5/2026, 7:43:00.388 AM	CSA303 - Resource Deletion in Lab Subscription - ASA-15643	CSA303 - Resource Deletion in Lab Subscription - ASA-15643

The scheduled analytics rule was successfully deployed and enabled in Microsoft Sentinel to monitor Azure resource deletion events within the CSA303 laboratory environment. Validation was performed by creating and deleting a temporary Network Security Group, which generated Azure Activity DELETE events. The activity was successfully ingested into the connected Log Analytics workspace and detected by the scheduled analytics rule, resulting in the generation of a Microsoft Sentinel alert. The measured detection delay between the successful deletion event and alert generation was approximately 10 minutes and 42 seconds, demonstrating the expected behaviour of Azure Activity log ingestion and scheduled analytics rule execution. This validation confirms that the detection logic, entity mapping, and incident generation configuration were functioning as intended for the laboratory environment.

Following the generation of the Microsoft Sentinel alert, the corresponding incident was opened for investigation. The incident was assigned to the analyst, and the investigation status was changed to Active. Alert details were reviewed without immediately assigning a final classification to ensure that all available evidence was examined before determining the nature of the activity.

The investigation focused on the alert metadata, associated entities, and Azure Activity logs to establish the sequence of events that triggered the detection.

Home > Microsoft Sentinel

Microsoft Sentinel | Incidents

Selected workspace: 'law-csa303-2026-asa-15643'

» + Create incident (Preview) Refresh Last 24 hours Actions Delete Security efficiency workbook Columns Guides & Feedback

1 Open incidents 1 New incidents 0 Active incidents

Open incidents by severity: High (0) Medium (1) Low (0) Informational (0)

Search by ID, title, tags, owner or product Severity: All Status: 2 selected Incident Provider name: All Alert product name: All Owner: All

Auto-refresh incidents

Severity	Incident number	Title	Alerts	Incident provider na...	Alert product name	Created time	Last update time	Owner	Status
Medium	1	CSA303 - Resource Deletion in Lab Su...	2	Microsoft Defender X...	Microsoft Sentinel	08/05/26, 08:43 AM	08/05/26, 11:18 AM	Theophilus Adeleke	New

Home > Microsoft Sentinel

Microsoft Sentinel | Incidents

Selected workspace: 'law-csa303-2026-asa-15643'

» + Create incident (Preview) Refresh Last 24 hours Actions Delete Security efficiency workbook Columns Guides & Feedback

1 Open incidents 0 New incidents 1 Active incidents

Open incidents by severity: High (0) Medium (1) Low (0) Informational (0)

Search by ID, title, tags, owner or product Severity: All Status: 2 selected Incident Provider name: All Alert product name: All Owner: All

Auto-refresh incidents

Severity	Incident number	Title	Alerts	Incident provider na...	Alert product name	Created time	Last update time	Owner	Status
Medium	1	CSA303 - Resource Deletion in Lab Su...	2	Microsoft Defender X...	Microsoft Sentinel	08/05/26, 08:43 AM	08/05/26, 11:19 AM	Theophilus Adeleke	Active

Alert and Entity Investigation

The generated Microsoft Sentinel incident was investigated by reviewing the associated alert details and linked entities. The investigation focused on identifying the account that initiated the operation, the affected laboratory resource group, the Azure resource involved, and the Correlation ID associated with the deletion activity. These artefacts were used to validate that the detected event originated from an authorised laboratory exercise and to correlate the alert with the corresponding Azure Activity logs.

Results					
Chart					
Add bookmark					
☐	TimeGenerated [UTC]	Caller	ResourceGroup	CorrelationId	ActivityStatusValue
☐	> 8/5/2026, 7:32:18.234 AM	asa2615643@icdfa.edu.ng	RG-CSA303-ASA-15643	1e15ad36-a272-4993-913f-f6d...	Success
☐	8/5/2026, 7:32:12.565 AM	asa2615643@icdfa.edu.ng	RG-CSA303-ASA-15643	1e15ad36-a272-4993-913f-f6d...	Accept
	TimeGenerated [UTC]	2026-08-05T07:32:12.5653826Z			
	Caller	asa2615643@icdfa.edu.ng			
	ResourceGroup	RG-CSA303-ASA-15643			
	CorrelationId	1e15ad36-a272-4993-913f-f6d310efb8d8			
	ActivityStatusValue	Accept			
☐	8/5/2026, 7:32:12.393 AM	asa2615643@icdfa.edu.ng	RG-CSA303-ASA-15643	1e15ad36-a272-4993-913f-f6d...	Start
	TimeGenerated [UTC]	2026-08-05T07:32:12.3935538Z			

Incident Timeline

A chronological timeline of the investigation was created by correlating Microsoft Sentinel alert information with the corresponding Azure Activity logs. The sequence of events was reconstructed using the recorded timestamps, operation names, activity status values, and Correlation ID to demonstrate the complete lifecycle of the authorised resource deletion activity.

Time (UTC)	Activity	Status	Evidence Source
08:32:12.393	NSG Delete operation initiated	Start	Azure Activity
08:32:12.565	NSG Delete request accepted	Accept	Azure Activity
08:32:18.234	NSG successfully deleted	Success	Azure Activity
08:43:00.388	Scheduled analytics rule executed and generated Microsoft Sentinel alert	Alert Generated	Microsoft Sentinel

Case Notes and Incident Closure

Following the investigation, case notes were documented within the Microsoft Sentinel incident to record the observed activity, investigation findings, risk assessment, validation results, containment decision, and recommended improvement. After confirming that the activity originated from an authorised laboratory exercise, the incident was closed using the closest available benign classification while documenting the reason for closure.

(0)

Informational (0)

2)



CSA303 – Resource Deletion in Lab Subscription – AS...
Incident number 1
[Investigate in Microsoft Defender XDR](#)

 Theophilus Ad...
Owner

 Active
Status

 Medium
Severity

tags



Incident link

https://portal.azure.com/#asset/Microsoft_Azure_Security_Insig...

 Last comment

(Total: 1)

Observed: Azure Activity recorded the successful deletion of the temporary Network Security Group (nsg-csa303-delete-test) by asa2615643@icdfa.edu.ng within the laboratory resource group rg-csa303-asa-15643. Risk: Unauthorised deletion of Azure resources could disrupt services, remove critical infrastructure or destroy forensic evidence. Validation: Azure Activity logs confirmed the operation through

Write a comment...



Case Comment

Observed: Azure Activity recorded the authorised deletion of nsg-csa303-delete-test by asa2615643@icdfa.edu.ng within the resource group rg-csa303-asa-15643. The operation MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/DELETE completed successfully and generated a Microsoft Sentinel alert during the scheduled rule execution.

Context: The activity occurred within the dedicated CSA303 laboratory environment and was intentionally performed to validate the scheduled analytics rule.

Risk: In a production environment, unauthorised deletion of Azure resources could disrupt business services, affect availability, or remove forensic evidence required for incident investigation.

Validation: Investigation of the Azure Activity logs confirmed that the deletion operation was initiated by the authorised laboratory account. The associated Correlation ID linked the Start, Accept, and Success events, confirming the expected Azure Resource Manager operation sequence.

Decision: No containment actions were required because the activity was authorised laboratory testing. The incident was closed after confirming successful validation of the detection rule.

Next Step / Improvement: In a production environment, approved maintenance activities should be supported by documented change records or temporary allowlists with a defined owner and expiry date rather than permanently excluding specific users from detection.

Incident Classification

Since this was an authorised lab activity, close the incident using the closest available option, such as:

- Classification: Benign Positive
- Classification Reason: Expected Activity (*or the closest equivalent available in your Sentinel portal*)
- Status: Closed

If your Sentinel portal doesn't have "Expected Activity", choose the closest available benign option and explain in your report that the activity was authorised testing.

Cleanup and Cost Verification

Laboratory Resource Cleanup

After all laboratory evidence had been collected and exported, the dedicated Azure resources created for the Microsoft Sentinel laboratory were removed to prevent unnecessary resource consumption and avoid additional charges. Resource cleanup was performed using the supplied cleanup script or by deleting the dedicated laboratory resource group, ensuring that only resources created for the laboratory exercise were removed.

The Azure Activity diagnostic configuration and shared Microsoft Sentinel components were left unchanged unless they had been created exclusively for the laboratory and authorised for removal by the instructor.

```
theophilus [ ~ ]$ ls
cleanup-sentinel-lab.sh
theophilus [ ~ ]$ chmod +x cleanup-sentinel-lab.sh

./cleanup-sentinel-lab.sh \
--student-id ASA-15643
This will delete only resource group: rg-csa303-asa-15643
Name                               Location
-----
rg-csa303-asa-15643 southafricanorth
Type DELETE to continue: DELETE
Deleted: rg-csa303-asa-15643
Separately disable/delete the lab analytics rule and close lab incidents. Remove connector policy only with instructor approval.
theophilus [ ~ ]$
```

Verification of Resource Removal

Following the cleanup process, the Azure CLI was used to verify that the dedicated laboratory resource group had been successfully removed. The verification was performed using the `az group exists` command, which checks whether the specified resource group is still present within the Azure subscription. A return value of `false` confirms that the resource group has been permanently deleted and that all associated laboratory resources have been removed.

```
Separately disable/delete the lab analytics rule and close lab incidents. Remove c
theophilus [ ~ ]$ az group exists --name rg-csa303-asa-15643
false
theophilus [ ~ ]$
```

Cost Verification

After the laboratory resources were removed, the Azure subscription was reviewed to verify whether any additional charges had been incurred during the Microsoft Sentinel laboratory exercise. The review was conducted to confirm that all billable resources had been successfully removed and that no unexpected costs remained associated with the laboratory deployment.

The cost verification confirmed that no unexpected charges were observed following the cleanup of the laboratory resources. The Azure for Students subscription remained within its available credit, indicating that the laboratory activities did not result in any high additional cost after resource deletion.

Sign-out and Security Verification

Upon completion of the laboratory exercise, the Azure Cloud Shell session was terminated, and the Azure portal session was signed out to prevent unauthorised

access. A final verification was performed to ensure that no Azure CLI credentials, exported authentication files, or other sensitive laboratory artefacts remained on any shared computer used during the exercise.

The Azure Cloud Shell session was successfully terminated, and the Azure portal session was securely signed out. No Azure CLI credentials or authentication files were retained on shared systems following the completion of the laboratory. This ensured that the laboratory environment was left in a secure state and aligned with cybersecurity best practices for protecting cloud administrative access.

These sections are sufficiently professional and do not require screenshots unless your instructor explicitly requests them.

Findings and Analysis

The laboratory successfully demonstrated the deployment and use of Microsoft Sentinel for monitoring Azure management activities.

- Successfully deployed Microsoft Sentinel and connected it to a Log Analytics Workspace.
- Configured the Azure Activity data connector to collect Azure management events.
- Generated authorised Azure resource activities to produce security logs for analysis.
- Used Kusto Query Language (KQL) to investigate resource creation, updates, deletions, failed operations, and log ingestion delays.
- Created and enabled a scheduled analytics rule that successfully detected Azure resource deletion events.
- Generated and investigated a Microsoft Sentinel alert and incident based on the deletion of a temporary Network Security Group (NSG).

- Correlated related Azure Activity events using the Correlation ID to validate the complete operation sequence.
- Confirmed that the detected activity was an authorised laboratory exercise and appropriately classified it as a Benign Positive (Expected Activity).
- Successfully cleaned up all laboratory resources and verified that no unnecessary Azure resources remained after the exercise.

Challenges Encountered

During the laboratory exercise, the following challenges were encountered:

- Azure policy restrictions prevented the creation of some resources, resulting in permission-related errors during the activity generation script.
- Microsoft Sentinel log ingestion delay caused a noticeable gap between the Azure Activity event and the generation of the corresponding alert.
- Changes in the Microsoft Sentinel portal interface made it difficult to locate some investigation features, such as incident comments and detailed alert information.
- Resource creation errors required troubleshooting and re-running commands before the laboratory activities could proceed successfully.
- Limited visibility of investigation details within the incident page required pivoting to Azure Activity logs to obtain the Caller, Resource ID, Resource Group, and Correlation ID for analysis.
- Analytics rule execution timing required additional waiting time before alerts and incidents were generated and available for investigation.

Recommendations

Based on the findings from this laboratory, the following recommendations are proposed:

- Implement continuous monitoring of Azure Activity Logs to improve visibility into administrative actions.
- Develop and regularly review Microsoft Sentinel analytics rules to detect suspicious resource management activities.
- Correlate security events using identifiers such as Correlation ID to improve investigation accuracy.
- Establish change management procedures to distinguish authorised administrative activities from malicious actions.
- Regularly review and optimise alert rules to minimise false positives while maintaining effective threat detection.
- Periodically validate incident response procedures through testing and simulation to ensure readiness for real-world security incidents.

Conclusion

This laboratory successfully demonstrated the deployment and operation of Microsoft Sentinel as a cloud-native SIEM solution within the Microsoft Azure environment. The exercise provided practical experience in configuring Azure resources, collecting and analysing Azure Activity Logs using Kusto Query Language (KQL), creating scheduled analytics rules, and investigating security alerts and incidents.

The laboratory also reinforced key Security Operations Centre (SOC) concepts, including event correlation, incident triage, and incident documentation. Overall, the exercise enhanced practical skills in Azure security monitoring, threat detection, and incident response, providing a solid foundation for applying Microsoft Sentinel in real-world cloud security operations.